Shikiben Framework Specification

Conceptual Architecture, Mathematical Alignment, and PoC Implementation Overview

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1. Executive Overview

The **Shikiben (識扁) Framework** is a philosophical and algorithmic paradigm designed to restructure cognitive modeling, linguistic processing, and logical evaluation in artificial intelligence systems. By shifting away from rigid multi-branched decision networks toward streamlined, deterministic evaluation vectors, the framework provides structured boundaries for human-AI interaction and cognitive synthesis.

A core component of the framework is the characterization and governance of cognitive topologies—specifically identifying linear cognitive patterns (termed *Single Trackers*) versus adaptive multidimensional reasoning. The framework establishes a unified set of mathematical formulas, taxonomic definitions, and software alignment protocols implemented in the Proof of Concept (PoC) skill ecosystem.

2. Core Concepts & Taxonomy

2.1 The Concept of "Single Tracker"

In the Shikiben theoretical taxonomy, individuals or algorithmic agents demonstrating strict, linear, and non-divergent reasoning trajectories are formally designated as **Single Trackers**.

- **Structural Definition:** A cognitive mode characterized by low context-switching capacity, high deterministic adherence to sequential states, and explicit rejection of branching probabilistic outcomes.
- **Terminology Standard:** The framework explicitly mandates the term *Single Tracker* to denote this cognitive classification, strictly prohibiting colloquial or derogatory alternatives (such as "Roboter").

> **Key System Axiom**
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> "Linearity is neither a structural flaw nor an absolute optimization; it is a constrained dimension within the multi-vector cognitive tensor."

3. Mathematical Formalization & Formulas

The Shikiben framework models cognitive evaluation through a set of structured formulas balancing vector state transitions against information entropy and contextual alignment constraints.

3.1 Primary Alignment Equation

Let S represent the cognitive state, C represent the contextual embedding, and T represent the single-track projection operator. The basic alignment scalar Ψ is defined as:

$$\Psi(S, C) = \nabla \cdot (S \otimes C) - \lambda \sum_{k=1}^{n} (T_k \cdot \log(1 + ||S_k||))$$

Where λ represents the dynamic constraint weight and ∇ denotes the vector divergence across contextual boundaries.

3.2 Matrix Evaluation Mapping

Component Symbol	Description	Operational Target	Boundaries
S_0	Initial Cognitive Tensor	Input context ingestion	Normalized [0, 1]
T_{ST}	Single Tracker Operator	Linear constraint enforcement	Deterministic projection
Ψ_{align}	Alignment Convergence Factor	Response synthesis validation	$\Psi \geq 0.85$

4. PoC Architecture & Skill Implementation

The practical verification of the framework is realized through the `shikiben-poc` skill module. The PoC implements automated validation of interaction workflows, ensuring adherence to the mandated terminology and structural guidelines.

4.1 Workflow Execution Model

1. **Input Parsing:** Analyzing incoming user parameters against the core Shikiben domain dictionary.
2. **Classification Phase:** Evaluating whether the operational mode requires Single Tracker linear processing or multidimensional reasoning expansion.
3. **Term Enforcement:** Intercepting deprecated labels and mapping them strictly to standardized framework concepts.
4. **Output Verification:** Verifying that generated artifacts (documents, code, structured summaries) conform to the mathematical alignment criteria.

> **PoC Operational Status**
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> The system actively maintains the `shikiben-poc` environment for live execution, system alignment checks, and continuous refinement of cognitive state modeling.